

Software Requirements Specification (SRS)

ERP System for Academic, Marketing & Finance, and Administration & HR Modules

1. Introduction

Purpose: This document defines the requirements for the ERP system integrating Academic, Marketing & Finance, and Administration & HR modules.

Scope: The ERP will be modular, web-based (React + Vite frontend, Node/Express backend, PostgreSQL database), and provide secure role-based access.

Users: Admin, Staff, Students.

References: Large System Development Project Document.

2. Overall Description

The ERP is modular, with separate APIs and dashboards.

Operating Environment: Node.js backend, PostgreSQL DB, React frontend.

Constraints: Responsive design, GDPR compliance, JWT authentication.

3. System Features

General Features:

- Authentication & RBAC
- Database management (normalized, backups)
- REST APIs

Academic Module: CRUD courses, exams, transcripts, attendance tracking.

Finance & Marketing Module: Invoices, payments, expenses, campaigns, ROI dashboards.

Admin & HR Module: Employees, payroll, leave, performance, inventory.

4. Non-Functional Requirements

Performance: API response < 300ms, scalable with Docker.

Security: JWT, password hashing, role-based permissions.

Reliability: Backups, availability 99.5%.

Maintainability: Modular code, documented APIs, Docker deployment.

5. Appendices

Database Entities:

- User(id, email, passwordHash, role, name)
- Course(id, code, title, credits)
- Invoice(id, studentId, amount, status)
- Employee(id, name, position, salary, hiredAt)

API Examples:

- POST /api/auth/login
- GET /api/academic/courses

- POST /api/finance/invoices/:id/pay